

Security Best Practices for Platform Engineers

NetGuard Solutions - Cybersecurity Thought Leadership

Platform engineers are responsible for the infrastructure that powers entire enterprise networks, making them one of the most critical points of risk when security is not treated as a priority. According to Gartner, by 2025, 99% of cloud security failures will be the customer's responsibility, with misconfigurations accounting for the vast majority of incidents. The challenge is not a lack of security tools it's the complexity of managing permissions, network segmentation, and security policies at scale.

The Capital One breach is a well-known example of this risk. The incident occurred after a former AWS employee exploited a misconfigured firewall within the company's cloud environment. A single overlooked configuration was enough to expose millions of sensitive records.

Three practices that significantly reduce this risk

1. Make Segmentation the Default, Not the Exception

Avoid overly permissive security groups with unrestricted inbound traffic. Design every service as though it were exposed to the public internet from day one.

2. Enforce the Principle of Least Privilege

Organizations adopting Zero Trust architectures reduce the likelihood of identity-related breaches by more than ****30%****. Permissions should be reviewed continuously-not only when resources are initially provisioned.

3. Continuously Monitor Your Environment

Misconfigurations often remain undetected for 70% of their lifecycle before being exploited. Automated compliance checks, continuous auditing, and real-time alerts help minimize that window of exposure.

Network security at scale is not a one-time initiative-it's an ongoing operational discipline. NetGuard Solutions helps platform engineering teams embed security into their daily operations, transforming it from reactive firefighting into a sustainable security practice.

Sources

1. Fidelis Security. Cloud Misconfiguration: The #1 Cause of Data Breaches 2025 (citing Gartner analysis).
2. Huntress. 27 Biggest Data Breaches in History: Famous Examples.
3. DataStackHub. 50 Cloud Breach Statistics for 2025-2026.